INTRODUCTION TO HEMODYNAMIC ASSESSMENT IN THE CARDIAC CATHETERIZATION LABORATORY

CHAPTER 1

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"Look into any man's heart you please, and you will always find, in every one, at least one black spot which he has to keep concealed."

Pillars of Society, act III

Henrik Ibsen (1877)

Despite the exhortations of poets and philosophers, the heart is, after all, simply a pump. The ability to "look into any man's heart" with the goal of understanding the function of this mysterious organ thwarted early generations of scientists and physicians. It would not take long, however, for science to garner the tools needed to peer into the hearts of mankind. Many of the major functions of the cardiovascular system important to our understanding of health and disease states are based on mechanical processes. Cardiac chambers contract and relax, valves open and close, and blood ebbs and flows based on the elementary principles of hydraulics. Contrast this with most other organ systems that exploit complex cellular and biochemical processes to accomplish their designated functions. For example, the kidneys balance fluid and electrolytes and excrete waste via an elaborate cellular array; the liver, pancreas, and intestinal cells digest food and absorb nutrients through a series of complicated biochemical steps, and muscle cells exert their cumulative toil through the elegant dance of complex protein molecules. The latter secrets eluded physicians and scientists, until only recently, when highly sophisticated tools became available to reveal the intricate and minute processes.

Many of the mechanical processes inherent to cardiac physiology can be understood by measuring changes in blood pressure and blood flow; the term *hemodynamics* refers to this discipline. Numerous brilliant investigators over many years have applied the study of hemodynamics to collectively expand our knowledge of cardiovascular physiology in both normal and pathologic conditions. The lessons learned from these generations of researchers rapidly became assimilated into the contemporary practice of clinical cardiology. Currently, hemodynamics is considered indispensable to the clinician managing patients with cardiovascular disease and forms the foundation of invasive and interventional cardiology.

A BRIEF HISTORY OF HEMODYNAMIC ASSESSMENT

All-important human endeavors possess histories replete with colorful anecdotes and legendary characters. The saga of cardiac catheterization is no exception.

The practical measurement of hemodynamics in humans required several crucial developments. These included the invention of safe and reliable catheterization techniques to access and study the right and left sides of the heart, the ability to image catheter position, and the creation of devices to convert pressure changes into an interpretable graphic form.

Insertion of tubes into the bladders and rectums of living people and the blood vessels of cadavers has been achieved since primitive times.[1] The first cardiac catheterization and pressure measurement performed on a living animal is attributed to the English physiologist Stephen Hales early in the 1700s and reported in the book *Haemastaticks* in 1733. By accessing the internal jugular vein and carotid artery of a horse, Hales performed his experiments using a brass pipe as the catheter, connected by a flexible goose trachea to a long glass column of fluid. The pressure in the white mare's beating heart raised a column of fluid in the glass tube over 9 feet high.[1]

As early as 1844, the famous French physiologist Claude Bernard performed numerous animal cardiac catheterizations designed to examine the source of metabolic activity. Many prominent scientists theorized that "combustion" occurred in the lungs. Using a thermometer inserted in the carotid artery, Bernard[2] compared the temperature of blood in a living horse's left ventricle with blood in the right ventricle, accessed from the internal jugular vein, and showed slightly higher right-sided temperatures, indicating that metabolism occurred in the tissues, not in the lungs. Bernard[2] also appeared to be the first to record intracardiac pressure using an early pressure recording system connected to the end of a glass tube inserted into a dog's right ventricle.

Later in the 1800s, in an attempt to address the controversy regarding the nature and timing of the cardiac apex beat, the French veterinarian Jean Baptiste Auguste Chauveau and physician Étienne Jules Marey performed

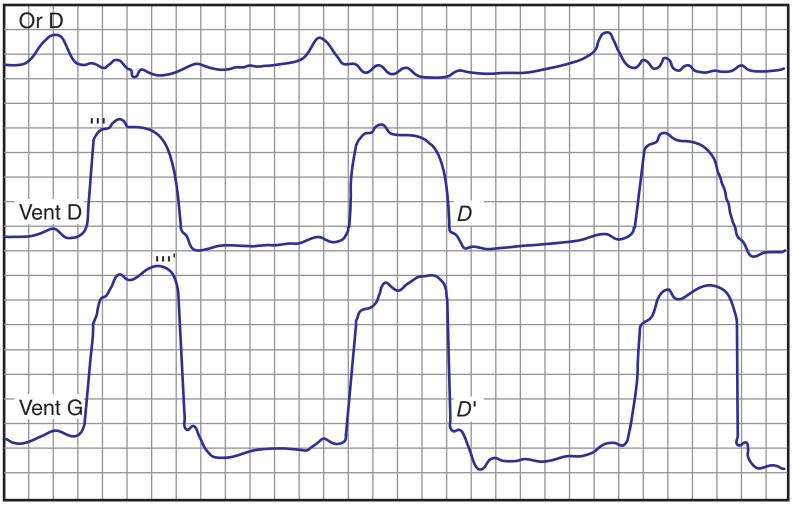

Fig. 1.1 Early pressure recordings were obtained from the cardiac chambers of a horse by Marey and Chauveau. (From Mueller RL, Sanborn TA. The history of interventional cardiology: cardiac catheterization, angioplasty, and related interventions. *Am Heart J*. 1995;129:146–172.)

catheterization using rubber catheters placed in a horse's jugular vein and carotid artery. These meticulous scientists recognized the importance of obtaining the highest quality data and recorded pressures in various cardiac chambers with clever mechanical devices invented by others but modified to suit their needs.[2] The graphic recordings obtained from these early transducers and physiologic recorders appear remarkably similar to those obtained in today's cardiac catheterization laboratories (Fig. 1.1).

From these early explorations of cardiac pressure measurement, there evolved an interest in quantifying blood flow. In 1870 the German mathematician and physiologist Adolph Fick published his famous formula for calculating cardiac output (oxygen consumption divided by arteriovenous oxygen difference).[3] However, Fick had more interest in the conceptual aspects of cardiac output determination than in its validation or application. The experiments necessary for validation of Fick's principle would fall to others more than 60 years later. Fick also contributed to the emerging field of hemodynamics with his valuable work refining early pressure recording devices.[3]

Despite numerous animal studies over many years, the placement of a catheter into the deep recesses of a living human heart would have to wait for an accurate method to image the course and position of the catheter. This would, ultimately, be feasible only after Wilhelm Roentgen's discovery of x-rays in 1895 (Fig. 1.2). The invention of an apparatus allowing us to peer inside the living human body for the first time represented one of the greatest medical advances in human history. At the start of the 20th century, it became possible to consider applying the lessons learned from animal research to humans. However, great trepidation remained among cardiovascular researchers because most considered the placement of a catheter into a living, beating human hearts foolhardy with potentially deadly consequences.

Although the historical record bestows acclaim for the first human cardiac catheterization to Werner Forssmann (performed on himself in 1929), his accomplishment may have been trumped by the little known, often disputed, and poorly documented efforts of fellow Germans Fritz Bleichroeder, E. Unger, and W. Loeb in 1905.[1,2] In an effort to deliver therapeutic injections close to the targeted organ, these physicians attempted to place catheters, without radiologic guidance, into the central venous circulation via the basilic and femoral veins. During one attempt made on his colleague Bleichroeder, Unger may have actually gotten into the heart because Bleichroeder reported the development of chest pain. They could not prove this theory because they failed to document the catheter position by x-ray or pressure recording and never published their observations, attempting to gain credit only after Forssmann received his in 1929.[1]

The account of Forssmann's first cardiac catheterization on himself, for which he was awarded the Nobel Prize in Medicine and Physiology in 1956, along with André Frederic Cournand and Dickinson Woodson Richards,[2,4–7] has been recounted numerous times and with several versions, some more engaging and colorful than others. The consistently told elements of his narrative are nearly unimaginable to contemporary physicians familiar with the existing training, medicolegal, and practice environments.

The essential facts of Forssmann's story are as follows. After graduating medical school, Forssmann began training as a surgical intern at the Auguste-Viktoria Hospital in Eberswalde, Germany, a small community hospital outside Berlin (Fig. 1.3). Forssmann's motivation in pursuing a means of instrumenting the right heart is unclear[6]; he reported that it evolved from the desire to find a method of infusing lifesaving drugs into the heart that was safer than direct

Fig. 1.2 Wilhelm Konrad Roentgen, discoverer of the x-ray. (From Edward P, Thompson D. *Roentgen Rays and Phenomena of the Anode and Cathode*. Van Nostrand Co.; 1896.)

Fig. 1.3 Werner Forssmann performed the first catheterization on himself at this hospital in Eberswalde, Germany. (From Forssmann-Falck R. Werner Forssmann: a pioneer of cardiology. *Am J Cardiol*. 1997;79:651–660.)

intramyocardial injection. Forssmann discussed his interest with his chief, Dr. Richard Schneider, but Schneider banned the enthusiastic intern from pursuing this work, largely because he thought it unlikely that mainstream German academic medicine would accept medical research from a community hospital. In addition, many considered the placement of a catheter into the heart very dangerous; Schneider did not wish notoriety for his hospital in the event that these investigations ended poorly.

Undeterred by the prevailing lack of support, Forssmann first placed catheters into the hearts of cadavers from an arm vein. Forssmann, impressed with the ease with which the catheters advanced, decided to perform the experiment on himself. As he was forbidden to proceed with any human experimentation by Schneider, he decided to carry out his project in secret. Forssmann recruited a colleague, Peter Romeis, and a surgical nurse, Gerda Ditzen, to assist

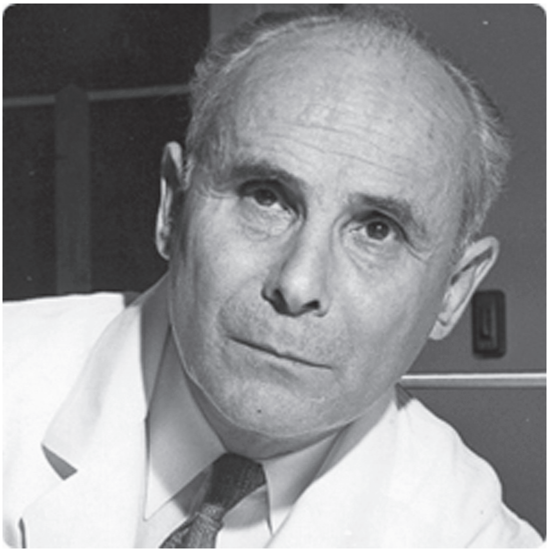

Fig. 1.4 André F. Cournand, MD, winner (along with Dickinson W. Richards and Werner Forssmann, not pictured) of the 1956 Nobel Prize in Medicine. (From Enson Y, Chamberlin MD. Cournand and Richards and the Bellevue Hospital Cardiopulmonary Laboratory. *Columbia Magazine.* Fall 2001: 34–44.)

him. Forssmann's first attempt failed. Peter Romeis performed the cutdown on Forssmann's cubital vein and advanced the catheter 35 cm, but Romeis lost courage, believing it too dangerous to continue, and stopped the experiment even though Forssmann felt fine. A week later, Forssmann chose a quiet afternoon when most of the hospital staff napped, and together with his nurse accomplice, he gathered the surgical instruments in an empty room to perform the procedure. Gerda Ditzen insisted on being the first participant and Forssmann played along, fully intending to perform the procedure on himself. After restraining the nurse to the table and preparing her incision site with iodine, Forssmann turned from her, quickly performed the venous cutdown on his own left arm, and inserted the ureteral catheter 65 cm. Ditzen became angry when she realized the deceit but quickly helped him walk down a corridor and two flights of stairs to the x-ray suite, where Forssmann confirmed the position of the catheter tip in his right atrium. Romeis apparently intercepted him in the x-ray suite to try to abort the experiment, but, according to one account, "… the only way Forssmann could hold him off was by kicking him in the shins."[4]

In his published account of his self-experimentation, Werner Forssmann[8] also describes a case where he used the catheter to deliver a solution of glucose, epinephrine, and strophanthin into the heart of a patient gravely ill with purulent peritonitis from a ruptured appendix. The patient died shortly after a brief period of improvement, and the autopsy confirmed the catheter position in the right atrium.

Forssmann's stunt did little to advance the field of cardiac catheterization beyond the bold demonstration that a catheter could actually be positioned safely in the human right atrium. No pressure measurements were made, and the catheter was not positioned in any other cardiac chambers. However, Forssmann had crossed the threshold and introduced the world to the potential of human cardiac catheterization.

Great turmoil and controversy followed Forssmann's publication. He failed to gain support from the medical community, and, while he continued investigations in cardiac catheterization (including at least six more self-experiments),[2] he became increasingly discouraged by the rigid, hierarchical nature of German academic medicine and became a urologist in private practice.

In the immediate years following Forssmann's success, a few isolated investigators dabbled in right-heart catheterization experiments.[1,2] However, nearly a decade would pass before there emerged a systematic discipline of right-heart catheterization exemplified by the classical work of André F. Cournand (Fig. 1.4) and Dickinson W. Richards at Columbia University's First Medical Division of Bellevue Hospital. The development of right-heart catheterization arose out of Cournand and Richards's interest in pulmonary function, measurement of blood flow, and the interactions between the heart and lungs in both health and disease. In the early 1930s, the group desired to measure pulmonary blood flow using the direct Fick method; however, this would require measuring mixed venous blood from the right heart, a feat considered too dangerous. Aware of Werner Forssmann's act, the group first demonstrated safety in animals and then placed modified urethral catheters in the right atrium of humans, sampling blood for oxygen content and making determinations of blood flow using Fick's principle. By the early 1940s a safe and valuable methodology of right-heart catheterization had been established, and Columbia became recognized as the first "cardiopulmonary laboratory" capable of applying these techniques to the study of cardiac and pulmonary diseases. With the onset of worldwide hostilities and imminent war, the group first directed their efforts to the analysis of blood flow in traumatic shock, making important observations valuable in wartime. After the war, Cournand, Richards, and others from their group published many landmark articles describing the hemodynamic findings in congenital heart disease, cor pulmonale, valvular heart disease, and pericardial disease. Much of our current understanding of these conditions has evolved from this important body of work.

Growing confidence and experience in right-heart catheterization techniques led to an interest in catheterization of the left heart. Catheter access to the left heart offered unique challenges and a much greater concern about safety, and initial adventures in accessing the left heart proved highly dangerous. Proposed and attempted methods to access the left ventricle included direct apical puncture, retrograde access from puncture of the thoracic or abdominal aorta, and a subxiphoid entry first into the right ventricle and then followed by puncture of the interventricular septum. Methods to directly access the left atrium included a transbronchial approach via a bronchoscope and a direct, posterior paravertebral left atrial puncture. It is interesting to note that reports of experiments involving self-catheterization similar to Werner Forssmann's involving the left heart are conspicuously absent from the literature.

Henry Zimmerman et al.[9] reported the first series of retrograde left-heart catheterizations from a left ulnar artery cutdown. This report noted failure to pass a catheter across the aortic valve from a retrograde approach in five normal patients, theorizing that the normal aortic valve prevented "against the stream" passage of the catheter, so they turned their attention to patients with aortic insufficiency. Zimmerman successfully entered the left ventricle in 11 patients with syphilitic aortic insufficiency. However, in a single patient with rheumatic aortic insufficiency, the attempt proved fatal. Present-day cardiologists engaged in the regular performance of left-heart catheterization would find their account shocking. While attempting to pass the catheter into the left ventricle:

> … *the subject suddenly complained of substernal chest pain and the electrocardiogram which was being recorded showed the abrupt appearance of ventricular fibrillation. The catheter was immediately withdrawn. Nine cubic centimeters of 1 percent solution of procaine with 0.5 cc of a 1:1000 solution of adrenalin were injected directly into the heart without effect on the cardiac mechanism. The heart was then exposed and massaged. This resulted in the restoration of a sinus rhythm, but the ventricular contractions were feeble and fifteen minutes after the onset of ventricular fibrillation the heart ceased beating..*[9]

With a failure rate of 100% in normal patients and an initial procedural mortality of nearly 10%, it is a wonder that further attempts at retrograde left-heart catheterization were made. However, perseverance improved the safety and success of retrograde left-heart catheterization to its currently recognized form. Additional advances included the development of transseptal catheterization techniques, simultaneous right- and left-heart catheterization, and, of course, angiography. By the end of the 1950s, right- and left-heart catheterization had become firmly established clinical techniques for the evaluation of valvular, structural, and congenital heart disease.

With most of the basic elements of catheterization techniques in place, investigators turned to refinement in equipment and techniques. Catheter design represented one of the first important refinements. The stiff, unwieldy catheters available to earlier generations of cardiovascular researchers required substantial manipulative skill to position and often caused significant arrhythmia. The invention of the balloon flotation catheter, exemplified by the Swan-Ganz catheter, represented the innovation leading to the universal acceptance and widespread practical application of hemodynamic assessment. The balloon flotation catheter became a clinical reality from the desire of Dr. Harold J.C. Swan, professor of medicine at the University of California, Los Angeles, and director of cardiology at Cedars-Sinai Medical Center, to apply cardiac catheterization techniques to study the physiology of acute myocardial infarction (Fig. 1.5). In the early 1960s cutting-edge hospitals began to develop specialized coronary care units to care for patients with acute myocardial infarction. Designed primarily to monitor and treat arrhythmias, coronary care units also became an obvious place to study the physiology of acute myocardial infarction. Early efforts to measure hemodynamics in potentially unstable patients with acute myocardial infarction using the stiff catheters and primitive techniques available at that time tended to induce life-threatening arrhythmias. Cardiologists considered catheterization dangerous during the acute phase of infarction and that it carried an unacceptable risk.

Swan became aware of the work of Ronald Bradley,[10] who reported the use of very small tubing to safely instrument the pulmonary artery and measure pressures in "severely ill" patients. When Swan attempted this technique, however, he found little success in passing the flimsy, small-caliber catheters from a peripheral vein to the pulmonary artery. In addition to the dearth of techniques to access a central vein, the most likely explanation for Swan's lack of success is related to the low output state of his patients compared to those of Bradley, preventing flotation of the catheter along the blood flow stream.

The answer to Swan's dilemma provides an entertaining and often-told example of the near-magical ability of the human mind to solve problems. Recalling this delightful story in his own words in 1991[11]:

> *In the fall of 1967, I had occasion to take my (then young) children to the beach in Santa Monica. On the previous evening, I had spent a frustrating hour with an extraordinary, pleasant but elderly lady in an unsuccessful attempt to place one of Bradley's catheters. It was a hot Saturday and the sailboats*

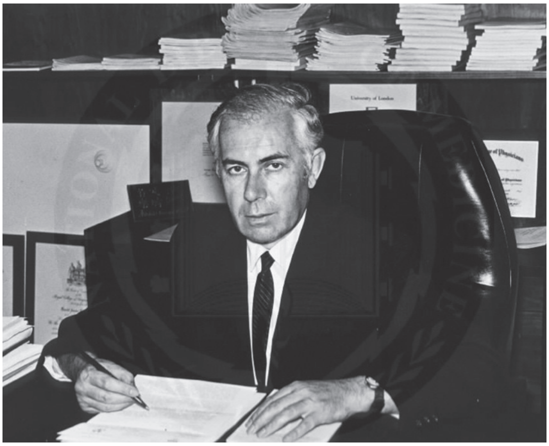

Fig. 1.5 Harold Swan, MD, codeveloper of the popular Swan-Ganz catheter. (Courtesy U.S. National Library of Medicine.)

on the water were becalmed. However, approximately half a mile offshore, I noted a boat with a large spinnaker well set and moving through the water at a reasonable velocity. The idea then came to put a sail or a parachute on the end of a highly flexible catheter and thereby increase the frequency of passage of the device into the pulmonary artery. I felt convinced that this approach would allow for rapid and safe placement of a flotation catheter without the use of fluoroscopy and would solve the problem of arrhythmias.

Edwards Laboratories worked with Swan to create the first five prototype catheters that relied on a balloon to accomplish flotation rather than parachutes or sails. (Interestingly, an early form of a balloon flotation catheter was described by Lategola and Rahn and failed to gain the attention of cardiovascular investigators; it did, however, prevent Swan from obtaining a patent on the idea.)[11]

Swan had previously hired William Ganz, an immigrant from the former country of Czechoslovakia and a survivor of the World War II labor camps, to work in the experimental laboratory at Cedars of Lebanon Hospital. The first animal experiments performed by Ganz with the prototype catheters were a brilliant success. Once the catheter was advanced into the right atrium and the balloon inflated, the catheter quickly migrated across the tricuspid valve and out of the pulmonary artery to the wedge position, confirming Swan's notion. The catheters were tried on humans with similar success and led to the landmark publication in the *New England Journal of Medicine*.[12] The group further refined the catheter's design, and Ganz added a thermistor to measure cardiac output using the thermodilution technique. Swan recognized that the catheter and procedure's success as a universally accepted bedside tool required that the technique be safe, easy to use, and would not interfere with routine nursing care in the intensive care unit. According to Swan[11]:

… *right heart catheterization became so routine and simple that the then Director of the Diagnostic Catheterization Laboratory, Dr. Harold Marcus, stated that he would ban the device because it was impossible to train the cardiac fellows in the appropriate manipulations of right heart catheters.*

The core elements of diagnostic cardiac catheterization and hemodynamic assessment have changed little since the 1970s. Innumerable additional contributors have refined catheterization techniques and expanded our knowledge of hemodynamics in health and disease; the valuable contributions of these notable leaders will be presented in subsequent chapters of this book.

While the bulk of attention is paid to the colorful pioneers of cardiac catheterization, the important role of the unglamorous physiologic recorder in the advancement of the science of hemodynamics is often ignored. In fact, the development of accurate physiologic recording equipment provided substantial challenges. The contributions made by mostly anonymous geniuses are easily forgotten but were as crucial to the development of cardiac catheterization as Roentgen's discovery of x-rays or Werner Forssmann's audacious self-experiments.

We take for granted the formidable task of translating a pressure wave sampled at the tip of the catheter to a graphic representation plotted as pressure versus time. The early pioneers of heart catheterization recorded intracardiac pressures in animals with primitive, mechanical contraptions consisting of elastic membranes attached to the catheter and using water-filled manometers that recorded pressure via a system of levers to a chart recorder (*sphygmograph*).[12] Springs and other clever mechanical adaptations to the devices improved their performance. Early

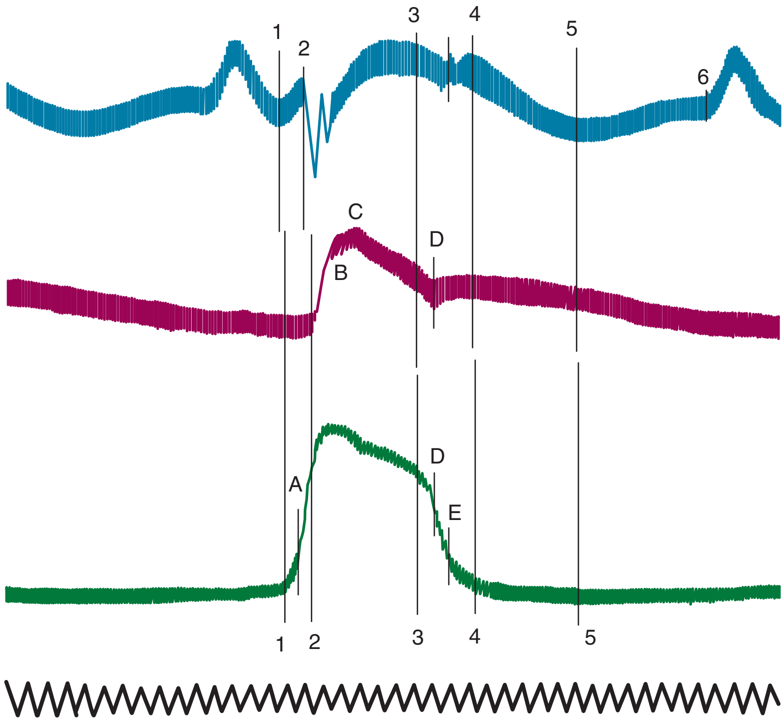

Fig. 1.6 High-fidelity recordings obtained by Carl Wiggers in 1921 from the right atrium (*top*), pulmonary artery (*middle*), and right ventricle (*bottom*) of a dog, using the optical manometer. (From Reeves JT. Carl J. Wiggers and the pulmonary circulation: a young man in search of excellence. *Am J Physiol*. 1998;274(4 Pt 1):L467–474.)

in the 20th century, several individuals made key contributions in this field. Carl J. Wiggers represents one of the key innovators in the development of high-fidelity pressure recording instruments.[13] He is credited with the invention of the Wiggers manometer, the first optical manometer. The optical manometer was designed based on the work originally conceptualized by Otto Frank. Wiggers spent time in Frank's Munich lab but was quite taken aback by Frank's secretive nature. Wiggers noted[13]:

> *Such a restrictive attitude in sharing newly developed apparatus was contrary to my scientific upbringing and threatened to frustrate my future use of them. Therefore, I connived with the laboratory mechanic who could use some extra money to make copies for me. In a sense, therefore, I smuggled the equipment I needed out of the laboratory.*

The configuration of the Wiggers optical manometer consisted of the catheter attached to a fluid-filled chamber. At the end of the small side arm of this chamber was an elastic membrane. A small mirror attached to this membrane reflected a light focused onto a light-sensitive recording paper. In this way, pressure changes from the catheter would be transmitted to the fluid-filled chamber and then to the membrane. The light beam essentially functioned as a weightless lever arm and a very sensitive method of reproducing rapid pressure changes. This innovation allowed the first high-fidelity measurements of intracardiac pressure (Fig. 1.6). Subsequent modifications by William F. Hamilton provided the essential equipment used in Cournand and Richards's laboratory at Bellevue.[14] Measuring and recording hemodynamics in that era required great patience and effort, as demonstrated in this description[14]:

> *Once the catheter was in place, all lights in the room were turned off, and the Hamilton manometer (which focused a light on sensitive paper to record the pressure contour) was attached to the catheter and manipulated in absolute darkness so that its light output could be captured with a handheld mirror and adjusted to strike the paper. Researchers could then record intravascular pressures.*

Advances in electronics changed the physiologic recorder. Oscilloscopes replaced the Hamilton manometer; the new systems converted catheter pressure to an electrical output displayed on cathode ray tubes. Some of us may still

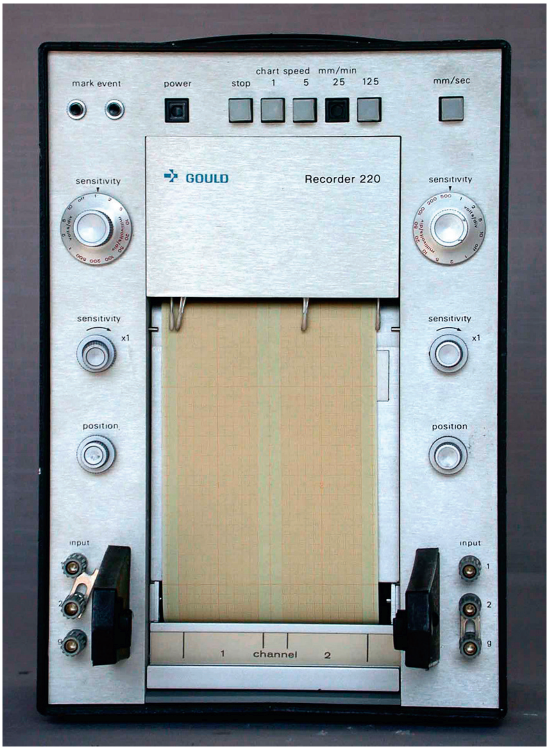

Fig. 1.7 Mechanical recorder used to collect hemodynamics and popular in the 1980s and early 1990s.

recall the old-fashioned chart recorders that used mechanical stylets to trace the pressure contour onto heat-sensitive paper for later analysis and storage (Fig. 1.7). These apparatuses have been replaced by tiny, cheap, and disposable table-mounted pressure transducers capable of converting a mechanical force to an electrical one, with subsequent conversion of this electrical signal in the “black box” of an advanced computer to the colorful graphic display that we have become accustomed to in the modern era cardiac catheterization laboratory (Fig. 1.8).

HEMODYNAMIC ASSESSMENT IN MODERN CLINICAL PRACTICE

The ease with which we can now assess cardiovascular hemodynamics has established cardiac catheterization as a routine diagnostic procedure. Furthermore, hemodynamic assessment was historically overshadowed by sexier developments in coronary intervention but has recently achieved greater attention and respect with the explosion of percutaneous valvular interventions, as well as the recognition of the importance of hemodynamic assessment for the optimal management of cardiogenic shock, heart failure, and pulmonary hypertension. Nearly all of the thousands of cardiac catheterizations performed each day in the United States measure left ventricular and aortic pressures; nearly a third of these also include the assessment of right-heart pressures and cardiac output. Many additional patients undergo right-heart catheterization alone in the cardiac catheterization laboratory. Medical, surgical, and coronary intensive care units contribute innumerable additional right-heart catheterization procedures performed at the bedside in critically ill patients, and anesthesiologists rely on right-heart pressure monitoring during many cardiac and high-risk surgical procedures in the operating room. Thus hemodynamic assessment has become an integral and established part of the daily practices of cardiologists, pulmonologists, anesthesiologists, surgeons, and intensivists.

There are many indications for invasive hemodynamic assessment. For patients referred to the cardiac catheterization laboratory, right- and left-heart catheterization is often performed for the evaluation and management of heart failure syndromes, shock, unexplained dyspnea, hypotension, respiratory failure, renal failure, edema, valvular heart disease, pericardial disease, hypertrophic cardiomyopathy, or congenital heart disease. Patients with unusual chest pain syndromes may require right-heart catheterization to exclude pulmonary hypertension. Most patients who undergo cardiac catheterization mainly for the evaluation of the coronary arteries, as seen in those with stable angina, abnormal stress tests, acute coronary syndromes, or uncomplicated myocardial infarction, require only measurement of left-heart

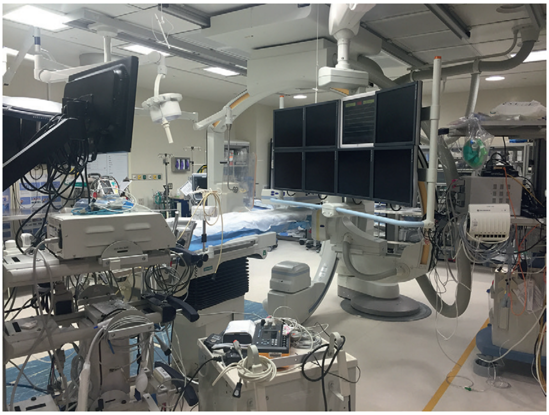

Fig. 1.8 The complex environment of the modern cardiac catheterization laboratory outfitted with computerized hemodynamic monitoring systems used by the University of Virginia Cardiac Catheterization Laboratories, c.2016.

and aortic pressure. However, postmyocardial infarction patients who exhibit hypotension, serious arrhythmia, or heart failure, or in the case of a suspected complication such as right ventricular infarction, ventricular septal defect, or mitral regurgitation should also undergo a careful right-heart catheterization. Patients under evaluation for heart or lung transplantation often undergo right-heart catheterization to identify pulmonary hypertension and, if present, a determination of reversibility by pharmacologic administration of a vasodilator agent.

Common indications for the bedside use of right-heart catheterization in patients with cardiac disease include the differentiation of cardiogenic from noncardiogenic causes of pulmonary edema, profound hypotension, or shock, and the guidance of therapy in patients with heart failure, pulmonary edema, pulmonary hypertension, or shock, particularly if there is renal impairment. The current recommendations on the indications and appropriate use of right-heart catheterization for both bedside and cardiac catheterization laboratory procedures have been described.[15–18]

EQUIPMENT

The essential components of a hemodynamic monitoring system include a catheter, a transducer, fluid-filled tubing to connect the catheter to the transducer, and a physiologic recorder to display, analyze, print, and store the hemodynamic waveforms generated.

A variety of catheters are available for pressure sampling (Fig. 1.9). The optimal catheter for hemodynamic measurements is stiff to transmit the pressure wave to the transducer without its absorption by the catheter, is easy and safe to position, and has a relatively large lumen opening to an end hole. The use of an end-hole catheter is especially important when sampling pressures within small chambers or when discerning pressure gradients over relatively small areas. However, an end-hole catheter may lead to damping or other artifacts if the end hole contacts the wall of the cardiac chamber. The commonly used “pigtail” catheter has multiple side holes and samples pressure at each of these openings, resulting in a tracing representing a mixture of the pressure waves collected at each opening. Such catheters are adequate for sampling pressure in a large, uniform chamber such as the aorta or left ventricle. It will not, however, have the required resolution to discern pressure gradients within the left ventricle. Catheters with an end hole and side holes at just the tip prevent damping or artifactual waveforms due to the positioning of the catheter tip against the chamber wall and are useful for collecting samples for oxygen saturation. The Swan-Ganz catheter is the most used catheter for measuring right-heart chamber pressures. In addition to the balloon at the tip for flotation, it consists of an end hole (distal port), a side hole 30 cm from the catheter tip (proximal port), and a thermistor for measurement of thermodilution cardiac output. This catheter is used extensively in modern cardiac catheterization laboratories as well as at the bedside for invasive monitoring. Other balloon flotation catheters include the Berman catheter, which is constructed of multiple side holes near the tip and no end hole or thermistor and is used principally for the performance of angiography, and the balloon-wedge catheter, which contains an end hole like the Swan-Ganz catheter but no thermistor for cardiac output measurement or additional infusion or pressure monitoring ports. Other catheters rarely used today for pressure measurement or blood sampling during right-heart catheterization do not use balloon flotation to assist in catheter positioning and must be directed carefully through the cardiac chambers under fluoroscopic guidance by the operator. These include the Layman catheter and Cournand catheter consisting of an end hole, the NIH catheter that contains multiple side holes near the tip but no end hole, and the Goodale-Lubin catheter consisting of an end hole and two single side holes near the tip used mostly for blood sampling.

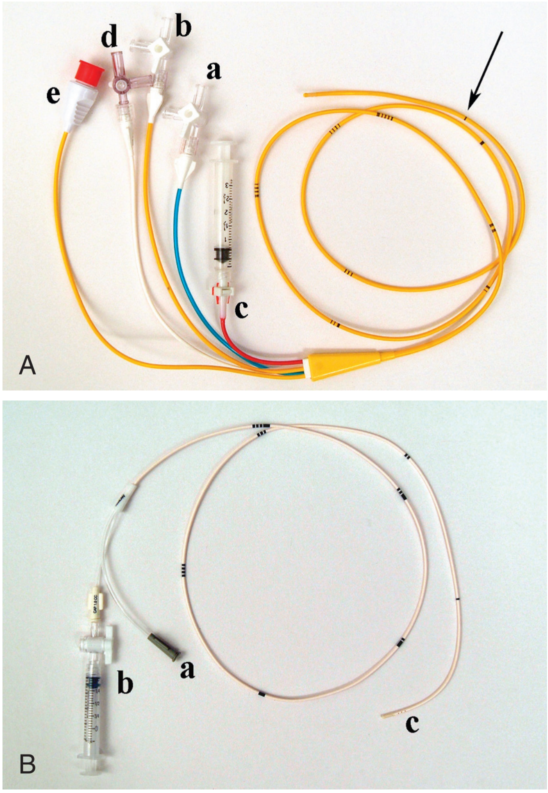

Fig. 1.9 Catheters are used for collecting hemodynamic measurements. (A) The popular Swan-Ganz catheter. This model has three ports consisting of a proximal lumen (*a*), a distal lumen (*b*), and the balloon port (*c*), which inflates the balloon mounted at the tip of the catheter. There is an extra infusion port (*d*) on this model. The thermistor for the performance of thermodilution cardiac outputs connects to the computer via a connecting plug (*e*). The catheter has 10-cm increments marked by lines (*arrow*). (B) Example of a Berman catheter. This is used not only for hemodynamics but also for angiography. There is a port connecting to the distal lumen (*a*) and a balloon inflation port (*b*). There are multiple side holes to allow angiography at the tip of the catheter (*c*).

Transducers and tubing constitute the next important component of the hemodynamic measurement system. Table-mounted, fluid-filled transducers currently used by most catheterization laboratories and intensive care units are inexpensive and disposable (Fig. 1.10). The pressure wave is transmitted through the fluid-filled catheter to a membrane in the transducer and deforms the membrane, resulting in a change in electrical resistance. This electrical signal is transmitted to the analyzing computer and converted to a graphic representation of the pressure wave. These relatively inexpensive transducers are factory calibrated but require “zeroing.” They sometimes do not hold calibration or a “zero” during use so should be replaced if suspicious or faulty data are obtained.

Fluid-filled systems are acceptable for clinical purposes but are subject to measuring artifacts. The catheter and connecting tubing should be stiff; soft tubing will absorb the pressure wave, damping and distorting it. In addition, the catheter as well as the tubing connecting the catheter and transducer should be as short as possible, with as few connections as possible to prevent timing delays, pressure damping, and a potential source of air bubbles. Great care should be taken to prevent kinking of the catheter or introducing air or a clot within the catheter or tubing because this will distort the waveform and lead to inaccuracies. Under certain circumstances, pressure may be measured directly in the cardiac chamber or vessel by use of a tiny transducer (micromanometer) mounted at the tip of a catheter, avoiding the limitations of a fluid-filled system. This is often the case when precise hemodynamic measurements are required as part of a research study but also form the basis of the pressure wire used for measurement of intracoronary pressure, which will be discussed in chapter 14.

Finally, a variety of proprietary computer systems are available for displaying, printing, and storing hemodynamic waveforms. The major systems in use are universally excellent and perform many of the analyses and calculations

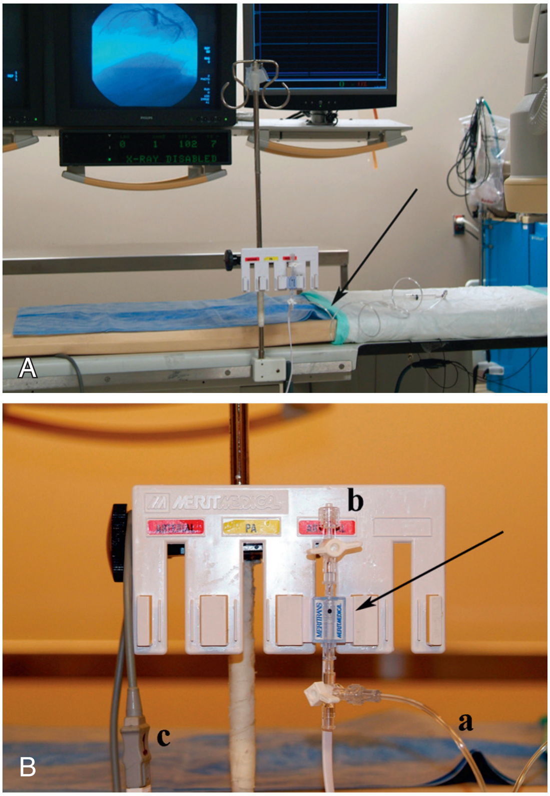

Fig. 1.10 Setup for a table-mounted transducer used for pressure measurement in the cardiac catheterization laboratory. (A) The general configuration. The catheter used to sample pressure is connected to high-pressure tubing (*arrow*). A close-up view of the transducer is shown in (B) (*arrow*). The high-pressure tubing connecting to the patient attaches by a stopcock to the transducer (*a*). Another stopcock allows flushing and equilibration with air (*b*). The transducer connects by cable to the hemodynamic computer (*c*).

Fig. 1.11 Modern hemodynamic systems automatically identify systolic (*s*) and diastolic (*d*) values and end-diastolic pressure (*e*) with a high degree of accuracy. In this example of a patient with aortic stenosis, the system automatically calculates the peak-to-peak and mean valve gradients and calculates the valve area.

previously done manually. These systems are quite good at analyzing the waveforms and automatically identify systolic and diastolic pressure points as well as "a" and "v" waves (Fig. 1.11). It is important to note, however, that the recognition algorithms in these systems occasionally misidentify waveforms, particularly if there is respiratory variation or artifact in the waveform or on the electrocardiogram (Fig. 1.12). For instance, left ventricular end-diastolic pressure or pulmonary artery systolic or diastolic pressure may not be properly identified if there is catheter whip or marked

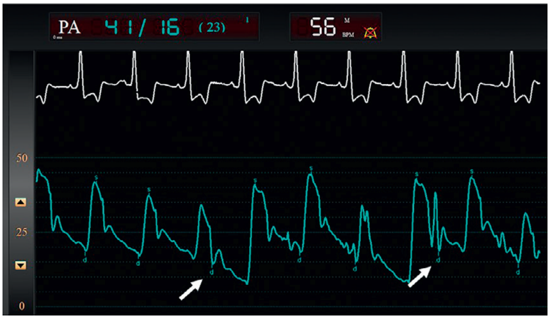

Fig. 1.12 Computer systems do not always identify the systolic or diastolic values accurately, particularly if there is an artifact or respiratory variation in the waveform. In this example, the pulmonary artery diastolic pressure is not correctly identified by the computer system (*arrows*).

respiratory variation. It is important for the operator to compare their own interpretation of the waveforms with the numbers provided by the computer to ensure accurate reporting of these values.

CATHETERIZATION PROTOCOLS FOR HEMODYNAMIC ASSESSMENT

The specific catheterization procedure and hemodynamic data collection protocol performed depend on the nature of the clinical question (Box 1.1). A right-heart catheterization alone is performed when the clinician needs to know the cardiac output, the right-sided chamber pressures, and the pulmonary artery pressure and resistance. If there is suspicion of an intracardiac left-to-right shunt, then a "saturation run" is necessary to determine the presence, size, and location of the shunt. A retrograde left-heart catheterization (i.e., catheter placed across the aortic valve and into the left ventricle) is used to measure the left ventricular diastolic pressure and the presence of an aortic valve or outflow tract pressure gradient. A transseptal left-heart catheterization is performed when it is desired to directly measure the left atrial pressure, usually to confirm the presence of mitral valve stenosis, or to measure left ventricular pressure in the presence of a mechanical aortic valve prosthesis. The most commonly performed and comprehensive hemodynamic procedure is a combined right-heart and retrograde left-heart catheterization. This is generally performed to assess heart failure syndromes and patients with presumed or confirmed valvular, congenital, pericardial, and myocardial diseases. Intracoronary hemodynamic measurements are specifically performed to assess the hemodynamic significance of ambiguous coronary artery lesions using a variety of methods based on the measurement of intracoronary pressure.

The protocol for right-heart catheterization is fairly simple. Venous access is obtained from the internal jugular, femoral, or brachial vein. Typically, a Swan-Ganz catheter is advanced to the right atrium, and the pressure in that chamber is recorded. With the balloon inflated, the catheter is advanced to the right ventricle, then the pulmonary artery, and finally to the "wedge" position with pressure waveform sampling obtained at each location. With the balloon deflated and the catheter tip in the pulmonary artery, a blood sample is obtained for measurement of oxygen saturation to screen for an intracardiac left-to-right shunt or to determine cardiac output using the Fick method. Thermodilution cardiac outputs are then measured with the catheter in this position.

During a complete right- and left-heart catheterization, the following routine is generally followed (Box 1.2). After obtaining arterial and venous access, the physician positions the Swan-Ganz catheter in the pulmonary artery and a pigtail catheter in the aorta. Thermodilution cardiac output is measured, and blood is sampled from the aorta and the pulmonary artery to calculate cardiac output using the Fick method and to screen for an intracardiac shunt. Aortic and pulmonary artery pressures are measured, and then the pigtail catheter is advanced in a retrograde fashion across the aortic valve and into the left ventricle. The Swan-Ganz catheter is advanced to the pulmonary capillary wedge position, and simultaneous left ventricular and pulmonary capillary wedge pressure is measured to screen for the presence of mitral stenosis. Pressure recordings are obtained from each chamber as the right-sided catheter is withdrawn with careful attention paid as the catheter crosses the pulmonic and tricuspid valves to screen for valvular lesions. Simultaneous right ventricular and left ventricular pressure recordings are obtained to screen for restrictive/constrictive physiology. Finally, careful observation of the pressure waveform, as the left ventricular catheter is pulled back into the aorta, serves as a screen for aortic valve stenosis.

As in most procedures, there are few "absolute" contraindications to cardiac catheterization. The risk-benefit ratio should be carefully considered in each individual. Relative contraindications for invasive hemodynamic assessment relate to patient features that increase procedural risk. While generally considered safe, there are multiple, serious potential complications from right- and left-heart catheterization (Box 1.3). The most observed complications relate to the access site, with hematoma, bleeding, and vessel injury not infrequent. Thus significant coagulopathy or thrombocytopenia or treatment with anticoagulant or thrombolytic drugs increases the risk of the procedure. Careful

Box 1.1 Indication for Specific Catheterization Procedures

- Right-heart catheterization cardiac output, right-sided chamber pressures
- Right-heart and saturation run left-to-right shunt determination
- Retrograde left-heart catheterization left ventricular pressure and gradients
- Transseptal left-heart catheterization direct measurement of left atrial pressure
- Right- and left-heart catheterization comprehensive hemodynamic assessment
- Intracoronary pressure measurement fractional flow reserve assessment

Box 1.2 Components of a Routine Complete Right- and Left-Heart Catheterization

1. Position pulmonary artery (PA) catheter.
2. Position aortic (AO) catheter.
3. Measure PA and AO pressure.
4. Measure thermodilution cardiac output.
5. Measure oxygen saturation in PA and AO blood samples to determine Fick output and screen for shunt.
6. Enter the left ventricle (LV) by retrograde crossing of the AO valve.
7. Advance PA catheter to the pulmonary capillary wedge position.
8. Measure simultaneous LV-pulmonary capillary wedge position.
9. Pull back from the pulmonary capillary wedge position to PA.
10. Pull back from PA to right ventricle (RV) to screen for pulmonic stenosis and record RV.
11. Record simultaneous LV-RV.
12. Pull back from RV to right atrium (RA) to screen for tricuspid stenosis and record RA.
13. Pull back from LV to AO to screen for aortic stenosis.

Box 1.3 Some Potential Risks of Right- and Left-Heart Catheterization

1. Access site complications
   - Bleeding
   - Hematoma
   - Vessel injury
   - Nerve injury
   - Infection
   - Pseudoaneurysm formation
   - Arteriovenous fistula formation
   - Pneumothorax (for internal jugular vein puncture)
   - Inadvertent arterial puncture
2. Arrhythmia
   - Ventricular tachycardia, ventricular fibrillation
   - Atrial arrhythmia, supraventricular tachycardia
   - Transient bundle branch block
   - Heart block
3. Myocardial infarction
4. Stroke
5. Infection
   - Bacteremia
   - Endocarditis
6. Pulmonary embolism/infarction
7. Pulmonary artery rupture
8. Vessel or cardiac chamber perforation
9. Catheter entrapment
10. Cholesterol embolization
11. Renal failure
12. Death

consideration should be given to patients with active infections, particularly bacteremia. Left-heart catheterization is frequently avoided in patients with a known left ventricular thrombus or active aortic valve endocarditis to minimize the risk of embolization. Arrhythmias are commonly seen during catheterization; most are due to catheter position, are transient, and of no clinical consequence. However, high-grade atrioventricular block can arise if the patient has underlying conduction abnormalities. Catheter placement in the right ventricular outflow tract can lead to right bundle branch block; left bundle branch block may arise when the aortic valve is crossed. Thus patients with an existing left bundle branch block may develop complete block when right-heart catheterization is performed; similarly, patients with underlying right bundle branch block may develop complete heart block when the catheter crosses the aortic valve during a left-heart catheterization. Normal conduction is generally restored with prompt removal of the offending catheter but may persist for some time and even require placement of a temporary pacemaker. Catheterization should be postponed, if possible, in patients with serious electrolyte or metabolic disarray because these may predispose the patient to the development of ventricular or atrial arrhythmias during catheterization.

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